

EXPERIMENT 18

AIM

Calculate SHA-256 values for files and group files having identical hashes.

ALGORITHM

1. Specify the evidence directory.
2. Create a SHA-256 function.
3. Calculate a digest for every regular file.
4. Group filenames by digest.
5. Check each hash group.
6. Classify groups with multiple files as duplicates.
7. Display the hash and filenames.
8. Stop.

CODE

```
import os
import hashlib
from collections import defaultdict

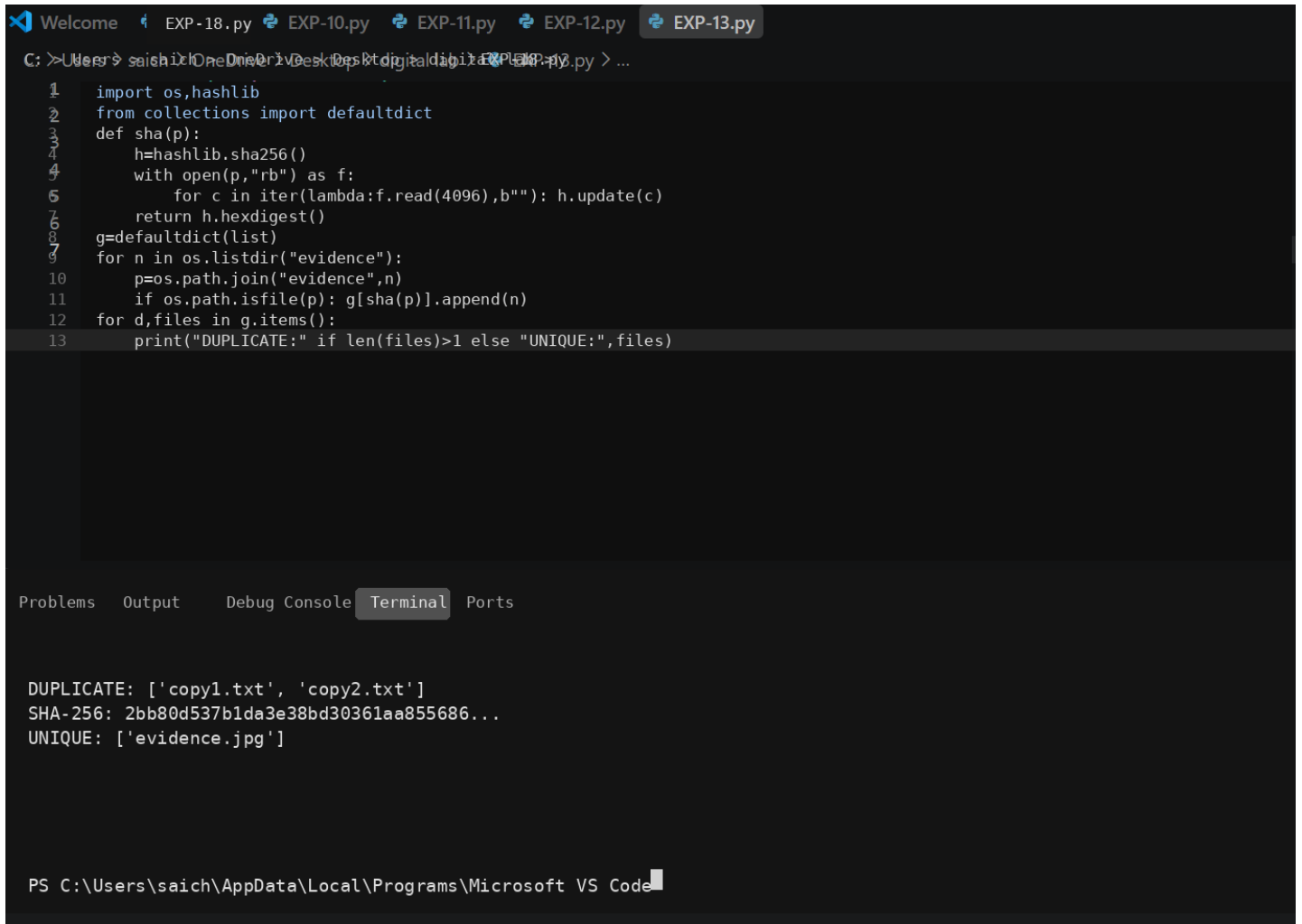
def sha256_file(path):
    digest = hashlib.sha256()
    with open(path, "rb") as file:
        for chunk in iter(lambda: file.read(4096), b''):
            digest.update(chunk)
    return digest.hexdigest()

groups = defaultdict(list)

for name in sorted(os.listdir("evidence")):
    path = os.path.join("evidence", name)
    if os.path.isfile(path):
        groups[sha256_file(path)].append(name)

for digest, files in groups.items():
    if len(files) > 1:
        print("DUPLICATE:", files)
    else:
        print("UNIQUE:", files)
    print("SHA-256:", digest)
```

OUTPUT



The screenshot shows a VS Code editor with a file explorer at the top displaying several Python files (EXP-18.py to EXP-13.py). The main editor window shows a Python script named EXP-13.py. The script imports `os` and `hashlib`, and uses `collections.defaultdict` to group files by their SHA-256 hashes. It iterates through the `evidence` directory, calculates the SHA-256 hash for each file, and appends the filename to a list in the dictionary. Finally, it prints the results, distinguishing between duplicate files (those with more than one entry) and unique files (those with only one entry).

```
1 import os,hashlib
2 from collections import defaultdict
3 def sha(p):
4     h=hashlib.sha256()
5     with open(p,"rb") as f:
6         for c in iter(lambda:f.read(4096),b''): h.update(c)
7     return h.hexdigest()
8 g=defaultdict(list)
9 for n in os.listdir("evidence"):
10     p=os.path.join("evidence",n)
11     if os.path.isfile(p): g[sha(p)].append(n)
12 for d,files in g.items():
13     print("DUPLICATE:" if len(files)>1 else "UNIQUE:",files)
```

The terminal output at the bottom shows the results of the script execution:

```
DUPLICATE: ['copy1.txt', 'copy2.txt']
SHA-256: 2bb80d537b1da3e38bd30361aa855686...
UNIQUE: ['evidence.jpg']
```

The status bar at the bottom indicates the file path: `PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code`.

RESULT

The program groups files with identical SHA-256 values to identify duplicate evidence copies.